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COMP 5212

Machine Learning

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Lecture 2: Hypothesis Sets and PAC Learning

Learning Goals

By the end of this lecture, you should be able to:

- define a **hypothesis set** and explain its inductive bias;
- distinguish **empirical** from **generalization** error;
- state the basic **PAC-learning guarantee**;
- derive the finite-class sample bound using a union bound;
- explain why infinite classes need a richer capacity measure.

Application: Build a Spam Filter

A new message arrives. Predict its label before it is seen.

Incoming message

Subject: FREE prize — claim now!

From: news@shop.example

contains “free” 1

contains “prize” 1

known sender 0

Learning task

- x : a feature vector describing the message;
- $y \in \{0, 1\}$: not spam or spam;
- S : labeled messages collected in the past.

What kinds of rules should the learner be allowed to use?

Data will choose a rule, but we choose the **menu of allowable rules** first.

One Data Set, Three Rule Families

The training messages are fixed; the menu of allowed rules is a modeling choice.

Keyword threshold

$$\mathcal{H}_1 = \{\text{score thresholds}\}$$

One score, one cut.
Simple and easy to estimate.

Conjunctions

$$\mathcal{H}_2 = \{\text{word conjunctions}\}$$

Example: “free” $\wedge$ “prize”.
Encodes a sparse, interpretable rule.

Memorization

$$\mathcal{H}_3 = \{h : \mathcal{X} \rightarrow \{0, 1\}\}$$

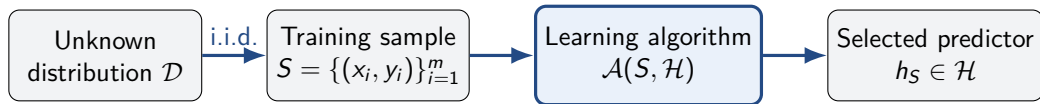
Can fit every observed message.
Flexible, but prone to overfitting.

The learner returns one $h_S \in \mathcal{H}$; the choice of $\mathcal{H}$ is the **inductive bias**.

Hypothesis Sets

What predictors may the learning algorithm choose?

From Data to a Predictor



The hypothesis set $\mathcal{H}$ is the learner's **menu of allowable predictors**.

Data select a predictor; the hypothesis set determines what can be selected.

Formal Setup

- $\mathcal{X}$: instance space.
- $\mathcal{Y} = \{0, 1\}$: label space.
- $c : \mathcal{X} \rightarrow \mathcal{Y}$: target concept.
- $\mathcal{C}$: possible target concepts.
- $\mathcal{D}$: unknown distribution on $\mathcal{X}$.

A labeled sample is

$$S = ((x_1, y_1), \dots, (x_m, y_m)), \quad y_i = c(x_i),$$

with

$$x_1, \dots, x_m \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}.$$

The learner sees S , but not c or $\mathcal{D}$.

Training and test inputs are assumed to follow the same $\mathcal{D}$.

What Is a Hypothesis Set?

A **hypothesis set** (or hypothesis class) is a collection

$$\mathcal{H} \subseteq \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$$

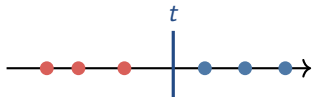
of candidate prediction rules available to the learner.

- The learning algorithm outputs $h_S \in \mathcal{H}$.
- $\mathcal{H}$ encodes an **inductive bias**: which explanations are considered plausible.
- Parameters index hypotheses; the entire parameterized family is the set $\mathcal{H}$.

Example: $\mathcal{H} = \{x \mapsto \mathbf{1}\{w^T x + b \geq 0\} : w \in \mathbb{R}^d, b \in \mathbb{R}\}$.

Examples of Hypothesis Sets

Thresholds on a line



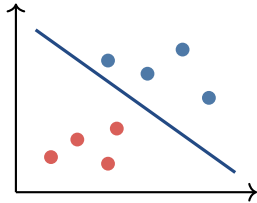
$$h_t(x) = \mathbf{1}\{x \geq t\}$$

Intervals



$$h_{a,b}(x) = \mathbf{1}\{a \leq x \leq b\}$$

Linear separators



$$h_{w,b}(x) = \mathbf{1}\{w^T x + b \geq 0\}$$

A hypothesis set may be finite or infinite; its capacity, not just its parameter count, controls generalization.

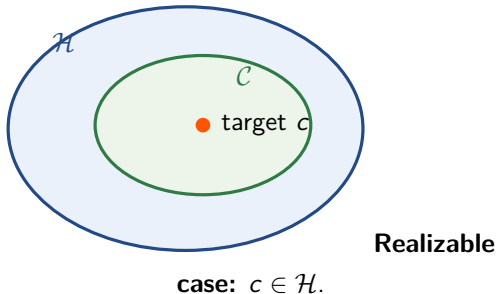
Concept Class vs. Hypothesis Set

Concept class $\mathcal{C}$

Describes possible data-generating target rules c .

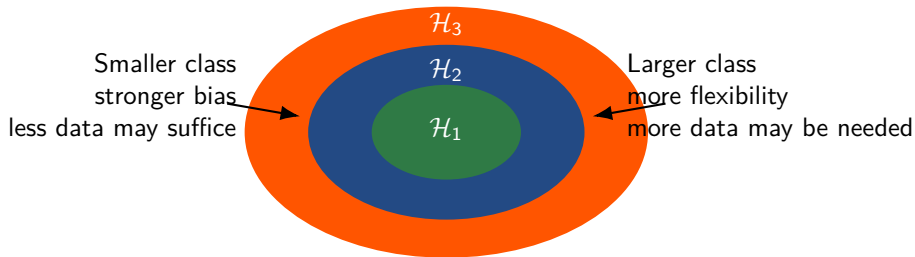
Hypothesis set $\mathcal{H}$

Describes predictors the algorithm is allowed to return.



If $c \notin \mathcal{H}$, the learner can only find the best approximation available inside $\mathcal{H}$; this is the **non-realizable** or **agnostic** setting.

Hypothesis Sets Encode Inductive Bias



Too small $\mathcal{H}$: approximation error and underfitting.

Too rich $\mathcal{H}$: more data-fitting explanations and overfitting risk.

Learning Rules: ERM and Consistency

A learning algorithm maps a sample to a hypothesis:

$$\mathcal{A} : S \longmapsto h_S \in \mathcal{H}.$$

Empirical risk minimization (ERM)

$$h_S \in \arg \min_{h \in \mathcal{H}} \hat{R}_S(h).$$

Choose the smallest training error.

Consistent learning

$$\hat{R}_S(h_S) = 0.$$

In a realizable problem, ERM can be consistent.

When does small training error imply small test error?

Empirical Error vs. Generalization Error

True error

$$R(h) = \Pr_{x \sim \mathcal{D}}[h(x) \neq c(x)].$$

Performance on future examples.

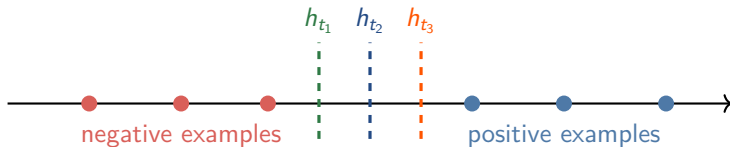
Empirical error

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{h(x_i) \neq c(x_i)\}.$$

Performance on the training sample.

For a fixed h , $\hat{R}_S(h)$ estimates $R(h)$. The subtlety is that the selected h_S depends on S .

Many Hypotheses Can Fit the Same Sample



The **version space** is

$$V_S = \{h \in \mathcal{H} : \hat{R}_S(h) = 0\}.$$

PAC analysis asks whether every member of V_S is likely to generalize.

The Realizable Case

The basic PAC model begins with a favorable assumption:

$$c \in \mathcal{H}.$$

Consequence

A zero-training-error hypothesis exists for every sample.

Question

How large must m be so that a consistent hypothesis has error at most ϵ ?

This leads to the Probably Approximately Correct framework.

PAC Learning

A distribution-free, finite-sample notion of learnability

PAC in Plain Language

Probably

With probability at least $1 - \delta$ over the random training sample.

Approximately

The returned predictor may not be perfect, but its error is at most ϵ .

Correct

Correctness is measured on new examples drawn from the unknown $\mathcal{D}$.

PAC learning turns “learns well” into a statement involving m , ϵ , and δ .

PAC-Learnability: Formal Definition

A concept class $\mathcal{C}$ is **PAC-learnable** if there exist a learner $\mathcal{A}$ and a polynomial sample bound $m_{\mathcal{C}}$ such that, for every $c \in \mathcal{C}$, distribution $\mathcal{D}$, and $\varepsilon, \delta \in (0, 1)$, whenever $m \geq m_{\mathcal{C}}(\varepsilon, \delta)$,

$$\Pr_{S \sim \mathcal{D}^m} [R(\mathcal{A}(S)) \leq \varepsilon] \geq 1 - \delta.$$

Universal

The guarantee holds for every target and every input distribution.

Probably

Probability is over the random training sample S .

Approximately correct

The returned error is at most ε .

Basic realizable PAC definition; Valiant (1984); Mohri et al. (2018), Chapter 2.

What the Basic PAC Model Assumes

Assumed

- examples are i.i.d.;
- training and test inputs share $\mathcal{D}$;
- the basic model is realizable: $c \in \mathcal{H}$;
- labels are binary in this presentation.

Not assumed

- a known parametric form for $\mathcal{D}$;
- Gaussian inputs or balanced classes;
- a particular learning algorithm;
- a specific geometry of $\mathcal{X}$.

PAC is **distribution-free**, but it does not cover arbitrary distribution shift.

How to Read ε and δ

Accuracy ε

We want

$$R(h_S) \leq \varepsilon.$$

Smaller ε means a more accurate predictor and usually requires more data.

Failure probability δ

We want the guarantee to hold with probability

$$1 - \delta.$$

Smaller δ means higher confidence and usually requires more data.

Example: $\varepsilon = 0.05$, $\delta = 0.01$ means error at most 5% with at least 99% confidence.

Sample Complexity

The **sample complexity** $m_{\mathcal{H}}(\varepsilon, \delta)$ is the number of labeled examples sufficient for a PAC guarantee over $\mathcal{H}$.

It records how data requirements scale with:

- desired accuracy $1/\varepsilon$;
- desired confidence $\ln(1/\delta)$;
- the complexity of $\mathcal{H}$;
- realizable versus agnostic assumptions.

Efficient PAC learning additionally requires polynomial running time.

Finite $\mathcal{H}$: Consistent Learning Bound

Assume:

- $\mathcal{H}$ is finite;
- $c \in \mathcal{H}$;
- the learner returns h_S with $\widehat{R}_S(h_S) = 0$.

With probability at least $1 - \delta$,

$$R(h_S) \leq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{m}.$$

Thus, to ensure $R(h_S) \leq \varepsilon$, it suffices that

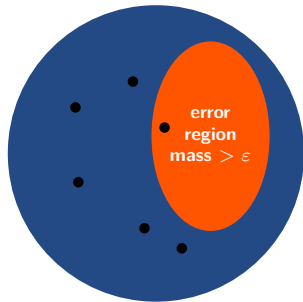
$$m \geq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{\varepsilon}.$$

Proof Idea I: A Bad Hypothesis Is Unlikely to Look Perfect

Fix h with $R(h) > \varepsilon$. Each example falls in the error region with probability greater than ε .

$$\Pr[\hat{R}_S(h) = 0] \leq (1 - \varepsilon)^m \leq e^{-m\varepsilon}.$$

A bad hypothesis can look perfect only if all m examples miss its error region.



Proof Idea II: Union Bound over All Bad Hypotheses

Define

$$\mathcal{H}_\varepsilon = \{h \in \mathcal{H} : R(h) > \varepsilon\}.$$

The learner fails only if some bad hypothesis is consistent:

$$\begin{aligned}\Pr\left[\exists h \in \mathcal{H}_\varepsilon : \hat{R}_S(h) = 0\right] &\leq \sum_{h \in \mathcal{H}_\varepsilon} \Pr\left[\hat{R}_S(h) = 0\right] \\ &\leq |\mathcal{H}|e^{-m\varepsilon}.\end{aligned}$$

Set $|\mathcal{H}|e^{-m\varepsilon} \leq \delta$ and solve:

$$m \geq \frac{\ln |\mathcal{H}| + \ln(1/\delta)}{\varepsilon}.$$

What the Finite-Class Bound Tells Us

Dependence	Interpretation
$1/\varepsilon$	Halving the target error roughly doubles the required sample size.
$\ln(1/\delta)$	High confidence costs only logarithmically more data.
$\ln \mathcal{H} $	The number of candidates enters through its logarithm.
consistency	The fast $1/m$ rate uses realizability and zero training error.

$\log_2 |\mathcal{H}|$ is the number of bits needed to identify a hypothesis: a formal version of **Occam's razor**.

Numerical Example: Discretized Thresholds

Suppose $|\mathcal{H}| = 1001$ candidate thresholds and we want $\varepsilon = 0.05$, $\delta = 0.05$.

$$m \geq \frac{\ln(1001) + \ln(20)}{0.05} \approx 198.1.$$

Therefore $m = 199$ examples suffice under the theorem's assumptions.

This is a worst-case guarantee, not a prediction of the exact data needed in a particular dataset.

Example: Conjunctions of Boolean Literals

Let $x \in \{0, 1\}^n$. A conjunction can include each variable positively, negatively, or not at all:

$$h(x) = x_1 \wedge \neg x_4 \wedge x_7.$$

Thus there are at most

$$|\mathcal{H}| = 3^n$$

conjunctions, and consistent PAC learning needs

$$m \geq \frac{n \ln 3 + \ln(1/\delta)}{\epsilon}.$$

For $n = 20$, $\epsilon = 0.10$, and $\delta = 0.05$, the bound gives $m \geq 250$.

Beyond the Realizable Case

Noisy labels, model mismatch, and uniform convergence

When No Hypothesis Is Perfectly Consistent

Real data may violate $c \in \mathcal{H}$:

- labels can be noisy or stochastic;
- the chosen model family may be misspecified;
- several labels may be plausible for the same input.

In the **agnostic setting**, compare the learner to the best hypothesis in $\mathcal{H}$:

$$h^* \in \arg \min_{h \in \mathcal{H}} R(h).$$

The goal is small **excess risk**

$$R(h_S) - R(h^*).$$

We now need training and true errors to be close *uniformly* over $\mathcal{H}$.

Hoeffding's Inequality for a Fixed Hypothesis

For a fixed h , the 0–1 losses are independent Bernoulli random variables. Hoeffding's inequality

$$\Pr\left[\left|R(h) - \widehat{R}_S(h)\right| > \gamma\right] \leq 2e^{-2m\gamma^2}.$$

gives

Why this is not yet enough

The output h_S is **not fixed**; it depends on the same sample S used to compute $\widehat{R}_S(h_S)$.

We need one event on which the bound holds simultaneously for every $h \in \mathcal{H}$.

Uniform Convergence for Finite $\mathcal{H}$

Apply Hoeffding to every $h \in \mathcal{H}$ and use a union bound.

With probability at least $1 - \delta$,

$$\forall h \in \mathcal{H}, \quad \left| R(h) - \hat{R}_S(h) \right| \leq \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

Uniform convergence lets us compare all hypotheses using the same training sample.

ERM Guarantee in the Agnostic Setting

Let $\hat{h}$ minimize empirical risk, let h^* minimize true risk over $\mathcal{H}$, and define

$$\gamma_m = \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

On the uniform-convergence event,

$$R(\hat{h}) \leq \hat{R}_S(\hat{h}) + \gamma_m \leq \hat{R}_S(h^*) + \gamma_m \leq R(h^*) + 2\gamma_m.$$

$$R(\hat{h}) \leq \min_{h \in \mathcal{H}} R(h) + 2\sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2m}}.$$

Approximation and Estimation

Approximation error

Best achievable risk inside $\mathcal{H}$



Reduce by enlarging or redesigning $\mathcal{H}$.

Estimation error

Price of selecting from finite data



Reduce with more data or lower-capacity $\mathcal{H}$.

Model selection balances the two: a richer hypothesis set improves fit but increases the amount of evidence needed for reliable selection.

What About Infinite Hypothesis Sets?

Many useful classes are infinite:

- real-valued thresholds;
- intervals with real endpoints;
- linear separators in $\mathbb{R}^d$.

Problem

The finite-class term $\ln |\mathcal{H}|$ is no longer useful.

Measure how many distinct labelings $\mathcal{H}$ can realize on finite samples. This leads to **growth functions**, **VC dimension**, and **Rademacher complexity**.

Key Takeaways

- $\mathcal{H}$ is the learner's allowable predictor family and its inductive bias.
- PAC means $R(h_S) \leq \varepsilon$ with probability at least $1 - \delta$.
- Finite, realizable classes need $m = O((\ln |\mathcal{H}| + \ln(1/\delta))/\varepsilon)$ examples.
- In the agnostic case, ERM excess risk is $O(\sqrt{(\ln |\mathcal{H}| + \ln(1/\delta))/m})$.
- Infinite classes require capacity measures beyond cardinality, such as VC dimension or Rademacher complexity.

Check Your Understanding

1. If $|\mathcal{H}|$ doubles, how much does the finite realizable sample bound increase?
2. Why can a fixed- h Hoeffding bound not be applied directly to the ERM output h_S ?
3. Give an example with $c \notin \mathcal{H}$. What is the correct comparison target?
4. Which is more expensive in the consistent bound: dividing ε by 10, or dividing δ by 10?

Next: VC dimension and guarantees for infinite hypothesis sets.

References and Further Reading



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Original instructional summary; definitions and finite-class bounds follow Chapter 2 of Mohri et al.